

Standard Operating Procedure

Incoming Material Inspection & Acceptance Criteria

Acme Precision Manufacturing Inc. — Quality Management System (ISO 9001:2015)

1. Purpose

This procedure defines the method, responsibilities, and acceptance criteria for inspecting incoming raw materials and purchased components prior to release into production inventory. It applies to all materials received under purchase orders issued by the Procurement Department and is intended to prevent nonconforming material from entering the manufacturing process. This procedure supports compliance with ISO 9001:2015 clause 8.4 (Control of Externally Provided Processes, Products and Services) and customer-specific quality requirements flowed down through purchase order terms and the Supplier Quality Manual (SQM-001).

2. Scope

This SOP covers all incoming materials classified as Critical or Major per the Approved Supplier List (ASL), including: machined metal components, injection-molded plastic parts, electronic subassemblies, and raw polymer resin. Minor commodity items (e.g., standard fasteners from certified suppliers) are exempt per Section 7.2 and follow reduced sampling per ANSI/ASQ Z1.4. This procedure applies to all three manufacturing sites (Plant 1 - Dallas, TX; Plant 2 - Monterrey, MX; Plant 3 - Reno, NV) and is maintained centrally by the Corporate Quality Department, with site-specific work instructions referenced in Section 10.

3. Definitions

Term	Definition
AQL	Acceptable Quality Limit — the maximum percent defective that can be considered satisfactory as a process average
COA	Certificate of Analysis — supplier-issued document certifying material properties for a specific lot
MRB	Material Review Board — cross-functional team that disposes nonconforming material
SCAR	Supplier Corrective Action Request — formal request for root cause and corrective action from a supplier
CMM	Coordinate Measuring Machine — precision instrument used for dimensional inspection
AOI	Automated Optical Inspection — machine vision system used to detect solder/assembly defects
QMS	Quality Management System — the documented system governing quality processes (this SOP is part of it)

4. Responsibilities

Role	Responsibility
Receiving Clerk	Logs shipment, verifies packing slip against PO, routes to QC Hold area
Quality Inspector	Performs dimensional/visual inspection, records results in QMS, applies disposition tag
Quality Engineer	Reviews inspection data for CAPA trends, approves deviations and MRB decisions
Supplier Quality Manager	Issues Supplier Corrective Action Requests (SCAR) for repeat nonconformances
Production Supervisor	Confirms material disposition status before pulling parts to the production line
Document Control Coordinator	Maintains current revision of this SOP and associated inspection forms

5. Acceptance Criteria by Material Type

Material Type	Inspection Method	Specification	AQL / Sample Size
Machined Aluminum Housing (Part No. 4471-A)	CMM dimensional check	Critical dims +/-0.05 mm per drawing Rev. F	AQL 1.0, Level II (ANSI/ASQ Z1.4)
Injection-Molded Bracket (Part No. 2209-C)	Visual + go/no-go gauge	No flash, warp, or sink marks > 0.2 mm	AQL 2.5, Level I
Electronic Subassembly (PCBA-110)	Functional test + AOI	100% functional pass; no solder defects	100% inspection
Raw Polymer Resin (Lot-traceable)	COA review + melt flow test	MFI 12-18 g/10min per ASTM D1238	Per lot, 1 sample/drum
Stainless Steel Fastener (Hardware Kit HK-07)	Visual + sample torque test	Class 8.8 min, no visible corrosion	AQL 4.0, Level I (reduced sampling)
Anodized Panel (Part No. 5581-D)	Visual + coating thickness gauge	Coating 12-18 microns, no color mismatch	AQL 1.5, Level II

Note: Full specification limits are maintained in the Engineering Drawing Package (EDP) and Supplier Quality Agreement for each part number. The values above are summary excerpts and must not be used as the controlling specification for production release decisions.

6. Inspection Procedure

- 1 Receiving logs the shipment in the ERP system and verifies quantity against the packing slip and purchase order.
- 2 Material is placed in the designated QC Hold area and is not released to production or stores until disposition.
- 3 The Quality Inspector pulls a sample per the AQL/sample size defined in Section 5 and performs the specified inspection method.
- 4 Inspection results, including measured values and any deviations, are recorded in the QMS inspection record (Form QC-014-F01).
- 5 First-article inspection (FAI) is performed on the first shipment of any new part number or after any engineering change, per AS9102 format where applicable.
- 6 If all characteristics pass, the inspector applies a green 'Accepted' tag and releases the material to inventory.
- 7 If any characteristic fails, the inspector applies a red 'Rejected' tag and initiates a Material Review Board (MRB) record per SOP-QC-020.
- 8 Repeat nonconformances from the same supplier within a rolling 90-day period trigger a Supplier Corrective Action Request (SCAR) per SOP-SQ-005.
- 9 Inspection equipment used must have a current calibration sticker; expired or unverified equipment may not be used (see Section 8).

7. Sampling Plan Reference (ANSI/ASQ Z1.4, Normal Inspection)

Lot Size Range	Sample Size (Level II)	Ac / Re (AQL 1.0)	Ac / Re (AQL 2.5)
2 - 8	2	0 / 1	0 / 1
91 - 150	20	1 / 2	2 / 3
151 - 280	32	2 / 3	3 / 4
281 - 500	50	3 / 4	5 / 6
501 - 1,200	80	5 / 6	7 / 8
1,201 - 3,200	125	7 / 8	10 / 11

Ac = Accept number (max defects allowed); Re = Reject number (defects at which lot is rejected). Switch to tightened inspection per Z1.4 switching rules after two rejected lots within five consecutive lots from the same supplier/part combination.

8. Inspection Equipment and Calibration

All measurement and test equipment used under this procedure (CMM probes, calipers, torque wrenches, AOI systems, melt flow indexers) shall be calibrated at intervals defined in the Equipment Calibration Master List (QA-REC-003) and traceable to NIST standards. Equipment due for calibration within 5 business days shall be flagged by the system and scheduled for service; equipment past due shall be immediately removed from service and tagged 'Out of Calibration.' Calibration records are retained for the life of the equipment plus 3 years.

Equipment Type	Calibration Interval	Standard/Method	Responsible Party
CMM (Coordinate Measuring Machine)	12 months	ISO 10360	External calibration lab
Digital Calipers / Micrometers	6 months	ASME B89.1.13	In-house metrology
Torque Wrenches	6 months	ASME B107.300	External calibration lab
Coating Thickness Gauge	12 months	ASTM B499	In-house metrology
AOI Vision System	Quarterly (verification)	Internal test board	QC Engineering

9. Nonconformance Disposition Options

Disposition	Description	Approval Required
Use-As-Is	Minor deviation does not affect form, fit, or function	Quality Engineer
Rework	Material can be brought into conformance via rework	Quality Engineer + Production
Return to Supplier	Material is non-repairable or out of spec	Supplier Quality Manager
Scrap	Material cannot be used, reworked, or returned	Quality Manager

10. Training Requirements

Personnel performing inspections under this SOP must complete initial training on this procedure, the applicable work instructions for their site (WI-QC-014-P1, -P2, -P3), and hands-on qualification on each piece of measurement equipment they will operate. Training records are maintained in the Learning Management System and refresher training is required every 24 months or upon any major revision to this SOP (revision level change of a letter or higher).

11. Records and Retention

Inspection records (Form QC-014-F01), Certificates of Analysis (COA), and MRB dispositions are retained for a minimum of 7 years in accordance with customer contractual requirements and ISO 9001:2015 clause 7.5.3. Records are stored in the QMS document control system and are subject to internal audit per the Annual Audit Schedule (SOP-QA-002). Where customer contracts specify a longer retention period, the longer period governs.

12. Referenced Documents

Document No.	Title
SOP-QC-020	Material Review Board (MRB) Process
SOP-SQ-005	Supplier Corrective Action Request (SCAR) Process
SOP-QA-002	Internal Audit Schedule and Process
SQM-001	Supplier Quality Manual
QA-REC-003	Equipment Calibration Master List
Form QC-014-F01	Incoming Inspection Record

13. Revision History

Rev	Date	Description of Change	Approved By
A	2023-06-01	Initial release	J. Alvarez, QA Director
B	2024-11-12	Added electronic subassembly AOI requirement	J. Alvarez, QA Director
C	2026-03-15	Updated AQL sampling table; added SCAR escalation criteria, Equipment Calibration section, and training requirements	M. Chen, QA Director

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